

Two-user egocentric QA generation pipeline

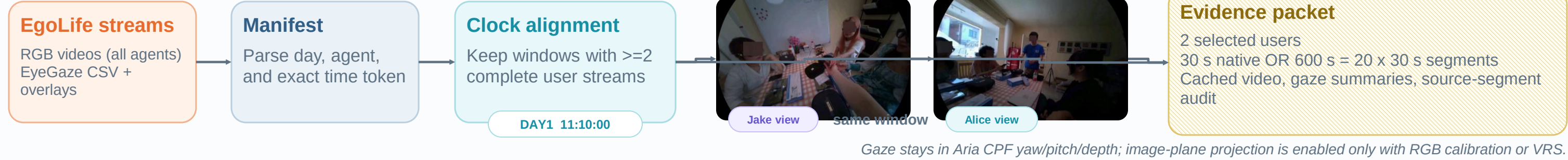
Synchronized streams -> CLIP-pruned generator media -> full-video verification -> auditable QA

BASELINE PRODUCTION

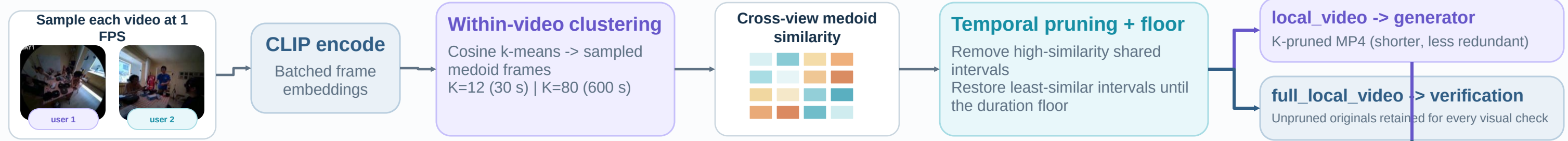
pruned media
full originals

retry feedback

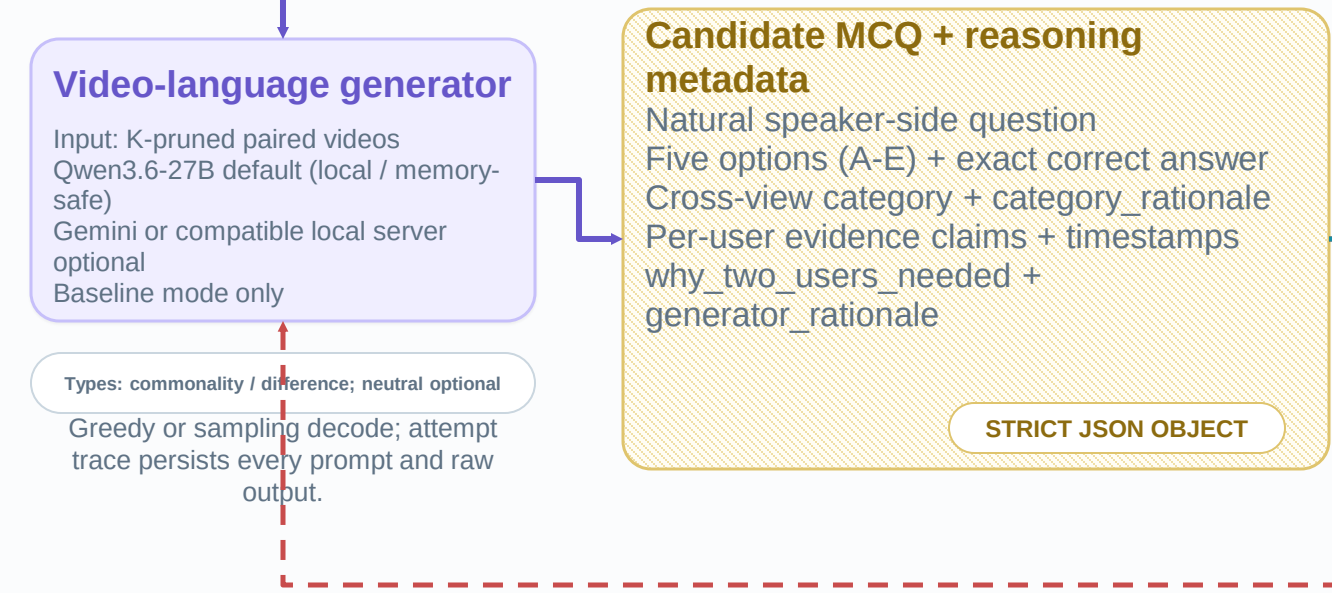
1 Build synchronized multi-user evidence



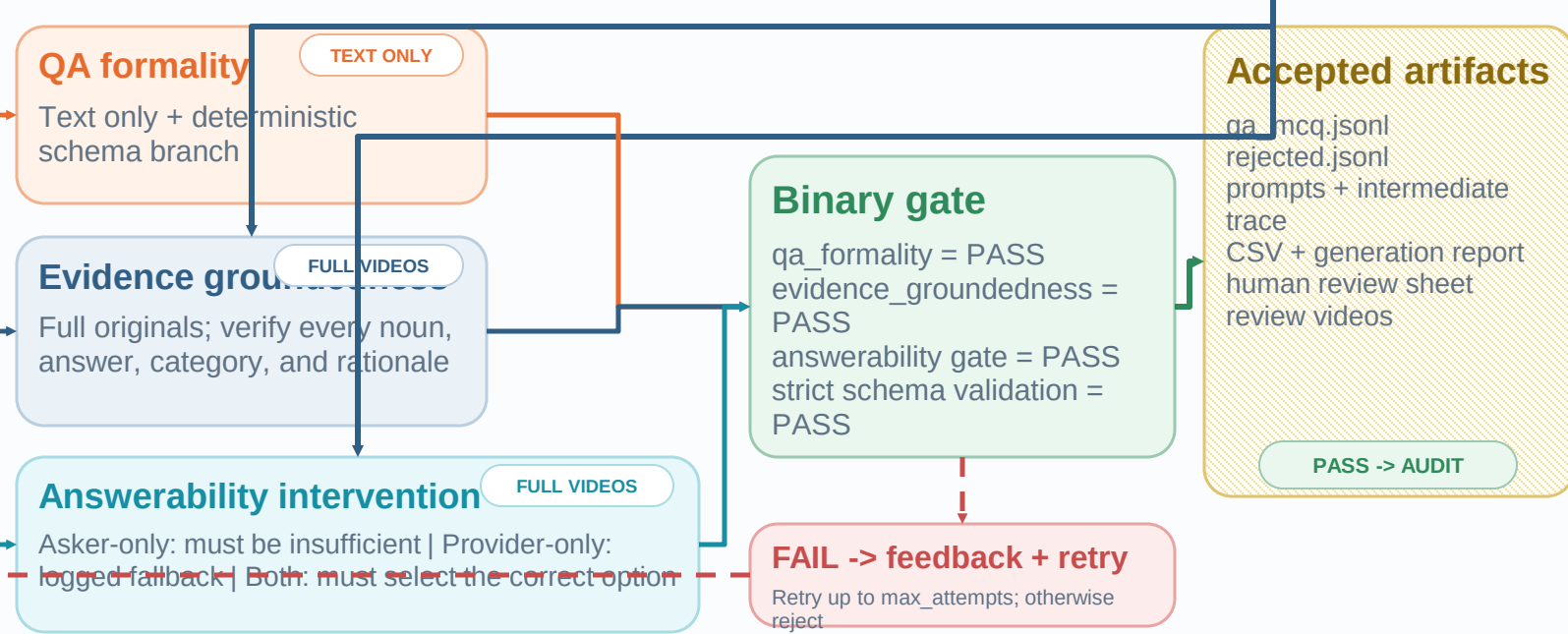
2 Prune redundant visual intervals and preserve full originals



3 Generate a situated two-user MCQ



4 Verify in parallel, gate, and retry



Audit and controlled experiments

Off the acceptance path: pruning K-grid with medoid/cluster traces + review.html | generator-rationale paired ablation | manual video review. Archived discovery / clip-guided modes and entropy scoring are not production gates.

Pruning illustration: what stays, what goes

Real fixed-pair K=12 run: DAY4 21:59:00, A2 Alice + A5 Katrina

K = 12

1 FPS

cosine >= 0.82

+/- 0.5 s interval

8 s keep floor

M1: 3 s <-> 26 s, cosine = 0.852

PRUNED

RESTORED

PRESERVED

VIDEO A - A2_ALICE

Full original: 30.0 s
Sampled frames: 30
Pruned: 21.5 s
Generator keeps: 8.5 s

FULL ORIGINAL RETAINED

0 s PRUNED

3 s PRUNED M1

5 s RESTORED

17 s PRESERVED

22 s PRESERVED

29 s PRESERVED

All 30 sampled frames

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

0s 5s 10s 15s 20s 25s 29s

Kept intervals: 4.5-5.5 | 15.5-18.5 | 21.5-22.5 | 25.5-27.5 | 28.5-30.0 s

8.5 s kept
21.5 s pruned

VIDEO B - A5_KATRINA

Full original: 30.0 s
Sampled frames: 30
Pruned: 21.5 s
Generator keeps: 8.5 s

FULL ORIGINAL RETAINED

0 s PRUNED

11 s RESTORED

16 s PRESERVED

20 s PRESERVED

23 s RESTORED

26 s PRUNED M1

All 30 sampled frames

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

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29

0s 5s 10s 15s 20s 25s 29s

Kept intervals: 10.5-11.5 | 14.5-17.5 | 19.5-20.5 | 21.5-24.5 | 29.5-30.0 s

8.5 s kept
21.5 s pruned

Generator route - local_video

Concatenate preserved + floor-restored intervals
Alice: 8.5 s | Katrina: 8.5 s
Temporal map links pruned time back to the originals

Verification route - full_local_video

Keep both unpruned 30 s originals unchanged
Evidence-groundedness + answerability always inspect full media
No pruned frame can hide an error from the judges

RESTORED = originally marked for removal, then reinserted from the least-similar candidates until the 8 s floor is met.